Branson Tay

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WORK EXPERIENCE

Jaguar TCS Racing **Aug. 2024 – Aug. 2025**
Junior Strategy & Software Engineer *Kidlington, United Kingdom*
Stack: Python, MATLAB, ReactJS, Typescript, CSS, MongoDB, Azure Pipelines, Docker, VBA

- Develop full-stack ReactJS apps visualising data critical to race strategy using live telemetry, tyre & timing data.
- Built and deployed a Python data pipeline converting Timing Feed data in NoSQL to processed ATLAS telemetry channels, following Agile practices, for performance analysis, tow analysis, and competitor analysis.
    - Implemented versioning, deployment pipeline and sustainable integration into existing systems.
- Deployed dockerised trackside tools via Azure DevOps with robust CI/CD, testing, and error handling.
- Wrote a custom Python library to extract and write car data to ATLAS format using MAT Atlas DLLs letting the Performance group engage in ML/AI projects such as virtual sensors, or large aggregations of car data.
- Troubleshoot and continuously improve existing codebases and race-critical tools live during race-weekends.
- Process pre-race battery simulations and simulator data results to monitor battery temperatures and ensure legality, correlating with track data to ensure accuracy in our models and to predict race temperature profiles.
- Logged and streamed live telemetry from WinTax using Visual Basic Scripts to a TCP socket via REST APIs.
- Perform qualitative competitor radio analysis (energy management, pit-stops, strategy, battery temperature management, vehicle setup and tyre run-plans) delivering concise per-session summaries to strategists.
- Analysed driver telemetry in ATLAS to optimise energy management and performance to gain race pace.

Republic of Singapore Navy **Nov. 2020 – Jul. 2022**
Marine Systems Specialist Technician *Singapore, Singapore*

- Trained on Singapore Navy's largest marine vessel, on maintenance and wartime awareness of ship systems.
- Led cross-team war-time firefighting simulation aboard ship engine operations room to assign appropriate fixes.
    - Analysed live sensor data in core ship systems from telemetry displays under high-pressure situations.
- Contributed to 50 pre- and post-mission debriefs, highlighting potential improvements for team efficiency.

Travelindr **Oct. 2019 – Mar. 2020**
Co-founder & CEO *Singapore, Singapore*

- Founded Travelindr, a group travel-itinerary generator, with a small team achieving a Minimal-Viable Product.
- Oversaw core strategies, product roadmap, data & competitor analysis, talent acquisition & market research.
- Conduct focus groups for qualitative data, pitched to investors, incubated in esteemed multi-national company.

EDUCATION

University of Leeds **Sep 2022. – Jul. 2026**
BSc Computer Science with Year in Industry
Leeds, UK

- **Formula Student Performance & Simulation Sub-team Lead**: Lead a diverse team of 12, with the Software Development Lifecycle and Agile, in projects like thermal management modelling, lift & coast simulators to quasi-static lap time simulators, ensuring the team follows a data-driven approach for design considerations.
- **Proposed Final Year Project**: Building a race strategy simulation platform using multi-agent reinforcement learning for the F1 2026 regulations. Models key race dynamics including tyre wear, overtaking difficulty, and energy deployment, investigating different algorithms and state spaces for optimal policy convergence.

Ngee Ann Polytechnic **Jan. 2017 - Mar. 2020**
Diploma in Mechanical Engineering *Singapore, Singapore*

- Utilised AutoCAD CAD software to fabricate gearbox given specific load, speed, and dimensional requirements
- Recognized with the Good Progress Award and the School of Engineering Merit Award.

TECHNICAL PROJECTS

Formula Student Lap Time Simulator |Python, Git, Vehicle Dynamics, Optimisation

- Developed quasi-static steady-state lap time simulators with Object-Oriented Programming using TTC tyre data, enabling sensitivity analysis of design parameters such as track width, wheelbase, and chassis stiffness.

2D CFD Formula Student Radiator Optimisation Simulator |Rust, Aerodynamics, Numerical Methods

- Developed a 2D incompressible fluid solver analysing radiator angle effects on cooling, pressure drop and drag.
- Automated batch simulations and visualisations to optimise radiator design, leveraging parallel processing.

Formula Student ICE Data Acquisition System|C++, Git, Arduino

- Aggregated data from GPS module to Arduino Teensy, visualising live data to the driver dash and over radio frequency for multiple departments to better quantify and enhance vehicle performance.
- Honed proficiency in collaborative software engineering tools like Git and conducting code reviews.

Formula Student EV Battery Management System|C++, CAN Bus protocol

- Spearheaded EV emulation of Battery Management System temperature sensor extension embedded system.
- Architected an expansion module using the CAN bus protocol for robust and efficient data transfer.
- Designed microprocessor framework to process battery module voltages to tangible data in C++.

Pathfinder Visualiser| ReactJS, Flask, HTML, CSS, JS

- Programmed a Full-Stack Web Application with Flask serving as a REST API and React as the frontend.
- Employed Dijkstra's algorithm and A* algorithm to visually demonstrate the speed difference between them.

ADDITIONAL EXPERIENCE

E-commerce Shopify Store Building | Facebook Ads Manager, Adobe Photoshop, Adobe Premiere Pro

- Developed and marketed multiple e-commerce shop brands by creating catchy, attention-grabbing creatives.
- Drove user attention by utilising good design principles like obvious action buttons, and consistent elements.
    - Increased conversions in by 10% by adapting "add-to-cart" button colour based on cursor heatmaps.
- Constructed efficient sales funnel, extracting data insights, and targeting trends, fully utilising data analysis tools.

Startup Partnership Intern*,* Global Entrepreneurial Internship Scholar

- Pioneered Voucher System project integrating cross-department efforts within a strict one-month window.
    - Achieved a 10% surge in the 30,000 annual customer base nationally.
- Independently secured over 10 successful strategic partnerships through presentations with senior executives.

TECHNICAL SKILLS, SOFT SKILLS & CERTIFICATIONS

Languages: Python, Matlab, HTML, CSS, Typescript, VBS, MongoQL, Rust, Java, C++, C.
Frameworks and Libraries: NumPy, SciPy, Bootstrap, Matplotlib, ReactJS, NodeJS, MongoDB, MySQL, Flask.
Tools: Docker, Git, Git Bash, Copilot, Azure DevOps, Azure Pipelines, Visual Studio, Microsoft Suite, ATLAS.

Soft Skills: Collaborative | Time-management | Curious | Adaptable | Multiperspectivity

Certifications & Awards

- Completed Harvard's introductory course to computer science, CS50.
- Runner-up in TRIVE Global Entrepreneurial Internship Program Startup Challenge.

HOBBIES & PERSONAL INTERESTS

- Engineering: Avid viewer of motorsport and airplane engineering videos|Karting.
- Stress-relieving: Padel|Table-tennis|Badminton|Cooking|Painting|Running.
- Enriching: Reading fiction & non-fiction books|Podcasts & videos on behavioural science.

Branson Tay

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Mercedes-AMG Petronas F1 Team
Lauda Drive
Brackley, Northamptonshire
United Kingdom

27th October 2025

Dear Hiring Manager & talent acquisition team,

I am an international student from Singapore pursuing a degree in computer science at the University of Leeds to fulfil my passion for solving complex engineering problems within my love of motorsport. My insatiable curiosity to understand how technical concepts turn into tangible solutions brings me exhilaration at the prospect of learning from and working with world-class engineers in the Mercedes AMG F1 team. There are three pivotal reasons why I am confident in my ability to excel as a Software Engineer.

Firstly, I excel at navigating ambiguity and delivering solutions under high-pressure scenarios. In Formula E Season 11 at Jaguar, new tyres and four-wheel drive for attack mode forced us to balance tyre warming with energy and battery management strategies. With engineers occupied by trackside preparations, I applied my mechanical engineering background and computer science data expertise to come up with an idea. I developed MATLAB scripts using simulator data to quantify per-lap friction brake energy needed to raise core tyre temperatures without excessive energy loss, later validating the approach with telemetry. Presenting these insights to engineers was a key step in holding off Porsche for P2 and securing a win in Shanghai Race 2. This experience highlights my adaptability to apply existing knowledge to evolving user requirements and to deliver high-performance solutions under pressure.

Secondly, I am a dedicated team player who knows when to work independently and when to seek input from others to achieve the best outcome. At Jaguar Racing, the compact team structure meant that responsibilities overlapped, requiring me to contribute simultaneously across simulations, software, and strategy. When developing a Python-based data pipeline converting raw Alkamel Timing Feed data from a noSQL database into telemetry channels, instead of developing in isolation, I sought feedback from the Vehicle Performance Group to ensure channels stayed relevant. I collaborated with software engineers to design the CI/CD architecture. I sought advice through code reviews with senior engineers for long-term scalability in object-oriented programming using design patterns, modularity, and extensive error handling. I also aligned with strategists to ensure the outputs had direct race impact. This balance of independence and collaboration turned what began as a single Python file into a fully functioning application still powering performance at Jaguar today. The experience reinforced my ability to thrive in multidisciplinary environments, translating user needs across the business into scalable software tools that allow for continuous improvement, an approach directly suited to being a Graduate Software Engineer in the Vehicle Performance Group.

Lastly, my curiosity to explore beyond my comfort zone drives my continuous growth. At Jaguar, I regularly engaged with colleagues across departments to understand their challenges, which led me to uncover an unused repository that once identified onboard clips from YouTube streams. Realising the potential, I set out to repurpose and extend the tool to include the ability to recognise dashes which contained key competitor data such as tyre pressures, which was needed by the tyre department. Despite having no prior experience in machine learning, I studied the fundamentals of the existing CNN classifier and sourced my own dataset. The innovation eased a major manual burden, freeing engineers to focus on performance-critical tasks. By taking initiative, I turned curiosity into measurable performance gains, a mindset directly suited for Formula One, within a project that enhanced existing systems while improving our understanding of our race car.

In this graduate programme, I aim to contribute from day one through my diverse knowledge from Formula E, mechanical engineering, and computer science to create high-performance solutions. I wish to surround myself with intelligent engineers with diverse skillsets to push my software engineering and communication skills in the early stage of my career. I am committed to leave no stone unturned in the chase for every millisecond for the Mercedes-AMG Petronas F1 Team.

I look forward to the chance to be a Graduate Software Engineer in Vehicle Performance, and I eagerly await your response.

Yours Faithfully,
Branson Tay